

Isla Stiven

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EDUCATION

Bachelor of Engineering in Mechanical Engineering (B.Eng)

University of Victoria | Thermo-fluids and Aerodynamics Specialization | GPA: 3.8/4.0

Expected Graduation: December 2026

Victoria, BC

SKILLS

Design Skills CAD | Design for Manufacturing | FEA | CFD | Rapid Prototyping | PCB Design | GD&T | Instrumentation | Engineering Drawing | Shop Tools | Embedded Systems | Fluid and Pneumatic Systems

Technologies SolidWorks | Siemens NX | ANSYS Granta EduPack | KiCAD | LabVIEW | MATLAB | Arduino | Python | C++

EXPERIENCE

ARTMS Inc.

Mechanical/Mechatronics Engineering Intern

January 2025 - August 2025

Burnaby, BC

- Owned the **design, build, and test** of a device used to measure internal component temperatures during **cyclotron irradiation**, engineered to meet strict requirements such as cyclotron interface compatibility, a fully sealed enclosure, and zero-interference with the irradiation process to improve understanding of **thermal failure modes**.
- Designed, assembled, and tested a **printed circuit board (PCB)** for ARTMS's **nickel-electroplating unit** used in the manufacturing of solid targets for **radioisotope production**.
- Employed **geometric dimensioning and tolerancing (GD&T)** on engineering drawings to define part tolerances, ensuring clear communication with **machinists**.
- Designed, conducted, and documented **experimental tests** that optimized isotope production, increasing **cooling flow 30%** through Swagelok-fitted water systems, cutting **production time 15%** via refined target-polishing methods, and mitigating system wear through **radiation-hard material selection**.
- Assembled, integrated, and tested** multiple major subsystems of the ARTMS **GEN2 QIS**, including **sensor/actuator** integration and fluid and **pneumatic systems**, deployed on cyclotrons to increase yield and improve efficiency of radioisotope production at **international sites**.

University of Victoria SEGAL Lab

Software Engineering Intern

September 2023 - December 2023

Victoria, BC

- Architected and built a cross-platform (iOS/Android) educational app prototype in **Flutter/Dart** with a 4-student start-up team, successfully delivering a functional MVP for teachers in remote communities, earning a **CBC interview** and an invite to the AISES National Conference.

PROJECTS AND INVOLVEMENT

Wingfoil Handle Force Measurement System

Tagmata Materials

January 2026 - Present

Victoria, BC

- Designed, simulated, built, and tested a functional **force measurement system** for wing foil athletes using **dual inline load cells** mounted to a **carbon fiber** auxiliary boom, enabling independent front and rear hand force capture during live water sessions to generate data for improving wing **aerodynamics and efficiency**.
- Developed a **sealed, IP67-rated electronics package** built around an **ESP32-S3** microcontroller, dual load-cell **ADC amplifiers**, a **9-DOF IMU**, and a **LiPo power system**, logging **time-synchronized force and orientation data** at **50 Hz** with **±5% FS accuracy** for reliable acquisition in saltwater environments.

UVic Formula Racing

Formula SAE Suspension and Accumulator Team Member

September 2024 - Present

Victoria, BC

Suspension A-Arm Node Design

- Designed four aluminum suspension A-arm nodes in **SolidWorks**, using **FEA, topology optimization**, and analytical hand calculations to achieve **64–70% mass reduction** per component while meeting safety factor requirements.
- Applied **design-for-manufacturing** principles to translate optimized geometries into **CNC-machinable** aluminum nodes bonded to **carbon fiber tubes**, sizing the adhesive joints to carry suspension loads simulated in Dynatune.

Accumulator Cooling System

- Optimized **accumulator cooling and waterproofing** by designing internal cooling ducts, external baffles, and fans based on **CFD results**, with **FSAE rules compliance**, sealing, and easy assembly/disassembly around the complex housing geometry as key design constraints.

The Victoria Hand Project

Volunteer Prosthetic Hand Technician

January 2023 - January 2024

Victoria, BC

- Fabricated and assembled functional upper-limb prosthetics from 3D-printed and laser-cut components, developing clearer assembly **documentation** to standardize the build process.

HOBBIES

Running, Hiking, Alpine and Nordic Skiing, Piano, Cycling, Volleyball